1 Introduction

The purpose of the Lab 3 is to provide understanding about how to build Flip-Flop circuits, how Flip-Flop circuits work without any external touch, role of the capacitors and transistors on the Flip-Flop circuit and how to build Oscillator-Timer Circuit (NE555 IC Time Machine) and how the resistance effects the frequency of the system.

2 Experiments

In this laboratory session we will held two different experiments. First experiment will be about the how to build and use Flip-Flop circuit while analyzing the circuit and second experiment will be about how to use Oscillator-Timer Circuit (NE555 IC Time Machine)

2.1 Experiment 1: Flip-Flops with capacitors and transistor

In our first experiment, we will construct a basic Flip-Flop circuit as shown in Figure 1. Before we delve into the experiment, that experiment will provide Flip-Flops are and how they work.

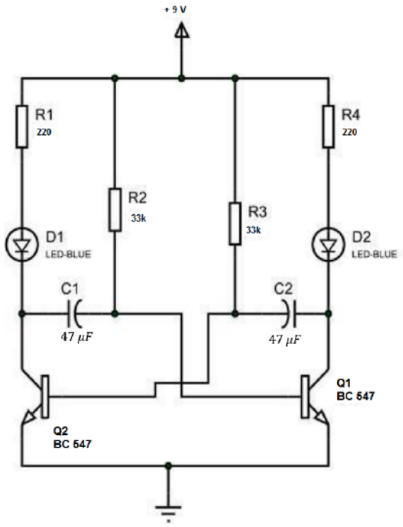


Figure 1: Flip-Flop circuits' schematic

In this experiment we used these components listed below.

- 1X Arduino Mega with USB cable to connect Arduino to computer
- 2X 220Ω resistor
- 2X 33kΩ resistor
- 2X 44μF capacitor
- 2X BC547 transistor
- 2X LED
- 2X Jumper cables
- 1X Breadboard

After we have built the circuit shown in the Figure 1, we observed the two different state of the circuit as shown in Figure 2 and Figure 3 below.

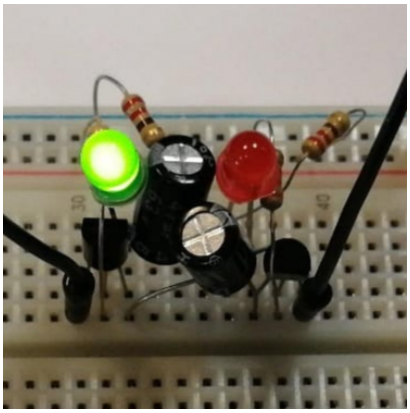


Figure 2: First state of the circuit

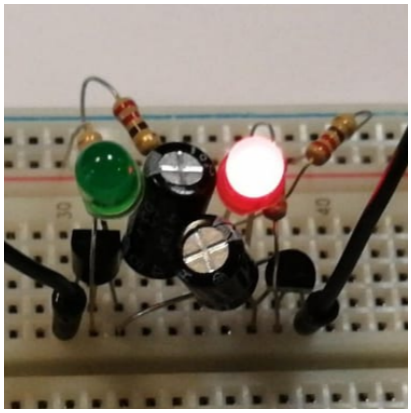


Figure 3: Second state of the circuit

As we can see from the figures above and the observations, one of the LEDs are lighting up everytime without any external help. When the circuit is powered ON, one of the transistors triggers first due to slight differences in the components or the power distribution. Then one of the transistor turns ON its corresponding LED. At the same time, it charges the capacitor connected to the base of the second transistor, keeping the second transistor OFF.

Once the capacitor is charged above the curtain voltage needed to activate the second transistor, the second transistor switches ON. This causes the first transistor to switch OFF, and the LED connected to the second transistor lights up while first capacitor drains.

2.2 Experiment 2: Oscillator-Timer Circuit (NE555 IC Time Machine)

In our second experiment, we will learn the workings of the 555 timer IC. This experiment will provide an understanding of the 555 timer IC and its various operating modes.

For the second experiment we have built the circuit shown in the Figure 4.

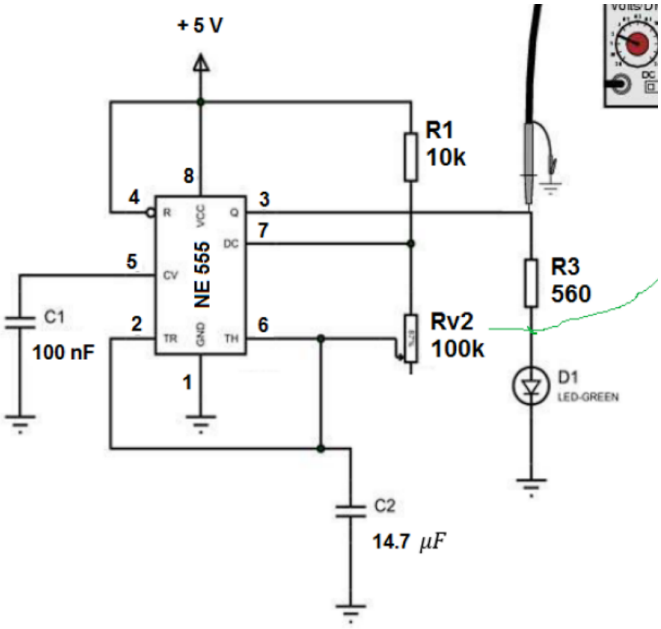


Figure 4: Oscillator-Timer schematic

In this experiment we used these components listed below.

- 1X NE555 IC
- 1X 10kΩ resistor
- 1X 560Ω resistor
- 1X 4.7μF capacitor
- 1X 10μF capacitor
- 1X 100nF capacitor
- 1X Potentiometer 100kΩ
- 1X LED
- 6X Jumper cables
- 1X Breadboard

After constructing the circuit for the second experiment we have the circuit shown in the Figure 5.

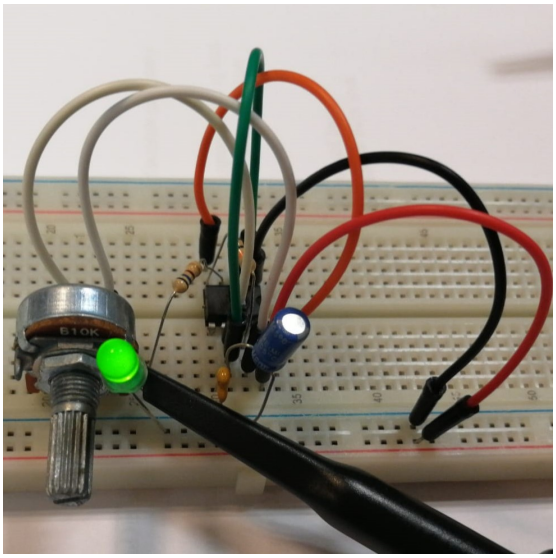


Figure 5: Oscillator-Timer schematic

We have connected oscilloscope to the one leg of the LED and captured the signal on the leg as shown in the Figure 5.



Figure 6: While potentiometer resistance is low

After measuring that signal from the circuit we have increased the resistance value of the potentiometer, and we have obtained the signals shown in the Figure 6. On the screen of oscilloscope we can see our signal frequency is 11.43Hz.



Figure 7: While potentiometer resistance is high

After increasing the resistance value of the potentiometer we have obtained that signal representation. As we can see from the Figure 7 we had frequency as 5Hz.

After our observation from the two different scenarios with different resistor values of potentiometer on the same circuit gives different frequency levels, we can also see that from the LED on the circuit which blinks with the frequency of the circuit.

So we can say if we increase the resistance level of the potentiometer system will work with lower frequency.

3 Conclusion

In this lab, we built and analyzed Flip-Flop circuits and a 555 timer IC circuit. We observed the bi-stable behavior of the Flip-Flop circuit, controlled by the charging and discharging of capacitors. In the 555 timer IC circuit, we found that the oscillation frequency can be adjusted by changing the resistance of a potentiometer. These experiments provided practical insights into the fundamental electronic components.